

First-Order Systems

Unit 4

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Input-output Models

■ General n^{th} order models

- input u , output y : these are signals or functions of time
- time is a continuous variable
- modeled by input-output differential equations

$$y^{[n]} = f\left(y^{[n-1]}, y^{[n-2]}, \dots, \dot{y}, y, u^{[m]}, u^{[m-1]}, \dots, \dot{u}, u\right)$$

- order = n or the highest derivative of y
- we require $n \geq m$ for causality

■ Linear Time Invariant Models

- focus of this class for now

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = b_mu^{(m)} + b_{m-1}u^{(m-1)} + \dots + b_1\dot{u} + b_0u$$

First-order Models

- Simplest possible model that has dynamics (i.e. derivatives)
- General first-order model

$$\dot{y} + ay = bu \text{ with initial conditions } y(0) = \xi$$

- here a, b are constants
- order = 1, relative degree = 1
- causal, linear, time-invariant, dynamic

- Example: car model with linear wind drag

$$m\dot{y} + \alpha y = Eu - Fd$$

y	car speed	m/sec
u	throttle motor voltage	volts
d	grade of roadway	percent

m	1000	Kg
α	60	Kg/sec
E	720	Kg m/ volt-sec ²
F	180	Kg-m/ sec ² -rad

The Total Response

- Consider a general first-order model

$$\dot{y} + ay = bu \text{ with initial conditions } y(0) = \xi$$

- General solution:

$$y(t) = \underbrace{e^{-at}\xi}_{y_h(t)} + \underbrace{\int_0^t e^{-a(t-\tau)} bu(\tau) d\tau}_{y_p(t)}$$

free response or homogeneous solution
due to initial cond ξ only with zero input $u(t) = 0$

forced response or particular solution
due to input u only with zero initial cond

$$y_h = e^{-at}\xi$$

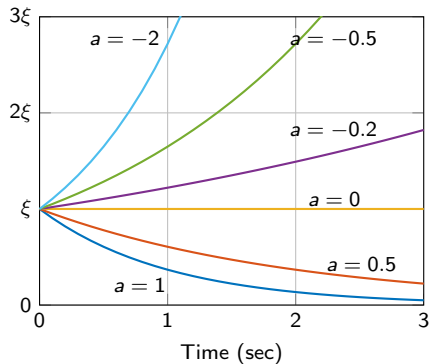
$$y_p = \int_0^t e^{-a(t-\tau)} bu(\tau) d\tau$$

Proof of Total Response Formula

- All we have to do is verify that the given solution
 - solves the differential equation
 - satisfies the initial cond
- Leibniz formula for derivative of an integral

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(t, \tau) d\tau = \frac{db(t)}{dt} \cdot f(t, b(t)) - \frac{da(t)}{dt} \cdot f(t, a(t)) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(t, \tau) d\tau$$

The Free Response



- free response decays to zero if and only if $a > 0$
- when $a < 0$ free response blows up
more negative $\rightarrow$ blows up faster
- when $a = 0$ free response stays constant
- when $a > 0$ free response decays to zero
more positive $\rightarrow$ decays faster

Stability

- Model is called **asymptotically stable** if free response decays to zero for all init condns

Stability of First Order Systems

First order system $\dot{y} + ay = bu$ is stable if and only if $a > 0$

- Many other notions of stability, but all are equivalent for LTI models

Time Constants

- Consider first-order model

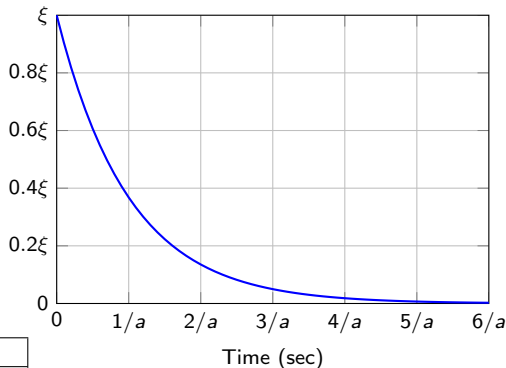
$$\dot{y} + ay = bu, \quad y(0) = \xi$$

- Assume stable, i.e. free response decays to zero, or $a > 0$

- Time constant $T = \frac{1}{a}$ seconds

captures decay rate of free response

- Free response has pretty much decayed in 3 time-constants

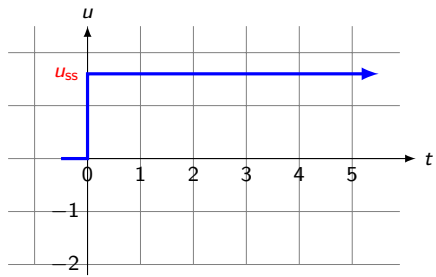


t	T	$2T$	$3T$	$4T$	$5T$
$y(t)$	0.37ξ	0.14ξ	0.05ξ	0.02ξ	0.0067ξ

BIBO Stability

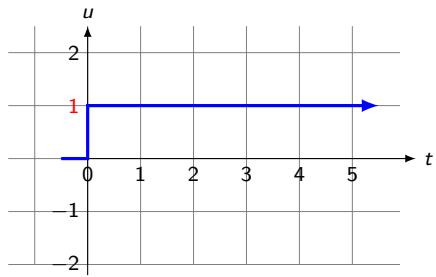
- Bounded signals
 - signal x is called bounded if $|x(t)| \leq M$ for all times t
 - $\cos(t)$ is bounded because $|\cos(t)| \leq 1$
 - e^t is not bounded
- Bounded input Bounded output stability
 - model $\mathcal{M}$ is BIBO stable if bounded inputs produce bounded outputs
 - yet another notion of stability

Step Inputs



General step input:

$$u(t) = \begin{cases} 0 & t < 0 \\ u_{ss} & t \geq 0 \end{cases}$$



Unit step input: we use the symbol μ

$$\mu(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

Step Response

- Consider first-order model $\dot{y} + ay = bu$
- Assume stable, i.e. free response decays to zero, or $a > 0$
- Response to general step input with init cond $y(0) = \xi$ is called the **step response**

$$\begin{aligned}
 y(t) &= e^{-at}y_o + \int_0^t e^{-a(t-\tau)} bu_{ss} d\tau = \underbrace{e^{-at}\xi}_{y_{\text{free}}(t)} + \underbrace{\frac{bu_{ss}}{a}(1 - e^{-at})}_{y_{\text{forced}}(t)} \\
 &= \underbrace{e^{-at}\xi - \frac{b}{a}u_{ss}e^{-at}}_{y_{\text{tr}}(t)} + \underbrace{\frac{b}{a}u_{ss}}_{y_{ss}(t)}
 \end{aligned}$$

- Response to unit step input with zero init conds is called the unit step response $g(t)$

Transient and Steady State Responses

- Total step response can be written as **free + forced = transient + steady state**

$$y(t) = \underbrace{e^{-at}\xi}_{y_{\text{free}}(t)} + \underbrace{\frac{bu_{\text{ss}}}{a}(1 - e^{-at})}_{y_{\text{forced}}(t)} = \underbrace{e^{-at}\xi - \frac{b}{a}u_{\text{ss}}e^{-at}}_{y_{\text{tr}}(t)} + \underbrace{\frac{b}{a}u_{\text{ss}}}_{y_{\text{ss}}(t)}$$

- Different decomposition: possible only because model is stable
- Transient response y_{tr} decays to zero
- Steady state response y_{ss} persists

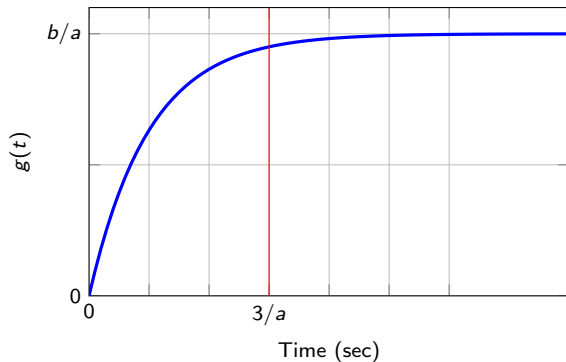
it is a constant term in the case of step inputs $y_{\text{ss}} = \frac{b}{a} \cdot u_{\text{ss}}$

- DC gain = $\frac{b}{a}$

Finding DC Gain

- Just set $\dot{y} = 0$, and calculate DC gain = $\frac{\text{output}}{\text{input}}$
 - ex: $2\dot{y} + 4y = u$ DC gain from u to $y = 1/4 = 0.25$
- Concept applies to any stable MIMO systems
 - pick any input u_i , and any output y_j
 - set all other inputs and outputs to zero
 - set $\dot{y}_j = 0$, and calculate DC gain = $\frac{y_j}{u_i}$
 - ex: $\dot{y} + 3y = r - 3d$ DC gain from r to $y = 1/3 = 0.33$, DC gain from d to $y = -1$

Unit Step Response



- initial value $y(0) = 0$
- final value $y(\infty) = \frac{b}{a} \cdot u_{ss}$
- exponential with time constant $T = 1/a$
- DC gain = $\frac{b}{a}$

Ex 1

Consider first-order system

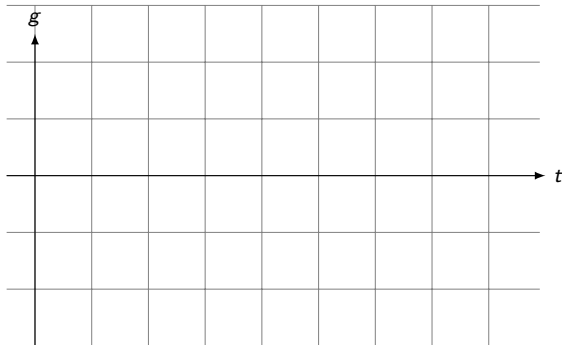
$$2\dot{y} + 4y = 2u \text{ with init condn } y(0) = -1$$

Find steady-state, transient, free, forced responses. Find DC gain and time-constant.

Ex 2

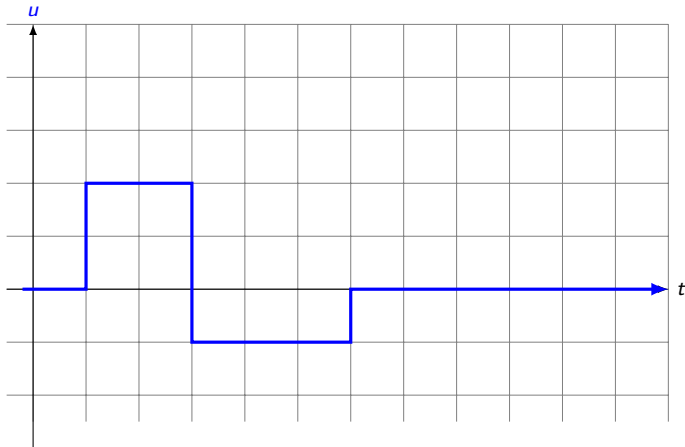
Easy to sketch step responses:

plot the unit step response of first-order system $\dot{y} + 10y = 20u$ with init condn $y(0) = -1$



Ex 3

Plot the response of the first-order system $\dot{y} + 1y = 2u$ with init condn $y(0) = 0$ to the input shown below



Design Example: Cruise Control

- First-order car model with linear wind drag

$$m\dot{y} + \alpha y = Eu - Fd$$

y	car speed	m/sec
u	throttle motor voltage	volts
d	grade of roadway	radians

m	1000	Kg
α	60	Kg/sec
E	720	Kg m/ volt-sec ²
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Proportional Control

- Proportional control strategy:

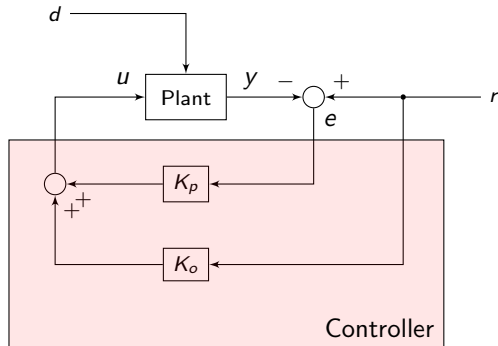
$$u = K_p(r - y) + K_o r$$

- Notice that

$$u = (K_p + K_o)r - K_p y = K_{ff}r + K_{fb}y$$

so this is equivalent to our old control scheme ...
 K_p is the feedback gain, K_o is the feed-forward gain

- Design: choose the gains K_p and K_o



Proportional Controller Design

■ Closed loop system dynamics:

$$m\dot{y} + \alpha y = EK_p \cdot (r - y) + EK_o \cdot r \quad \implies \quad m\dot{y} + (\alpha + EK_p)y = (EK_p + EK_o)r$$

■ Perfect tracking for model with no disturbances

- DC gain from r to $y = 1$
- set $\dot{y} = 0$ and $y = r$

$$(\alpha + EK_p)r = (EK_p + EK_o)r \quad \implies \quad K_o = \frac{\alpha}{E} = 0.0833$$

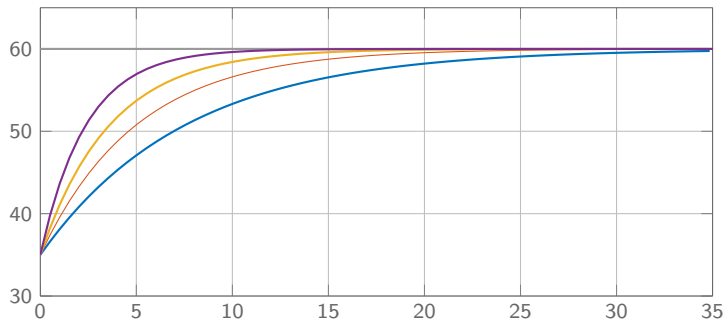
■ What can we do with K_p ? Change the time constant!

- suppose we want to reach $r = 65$ mph in 15 sec
- $3T = 15 \implies T = 5$ sec

$$T = 5 = \frac{1}{a} = \frac{1}{\alpha/m + EK_p/m} = \frac{m}{\alpha + EK_p} \quad \implies \quad K_p = \frac{m - 5\alpha}{5E} = 0.194$$

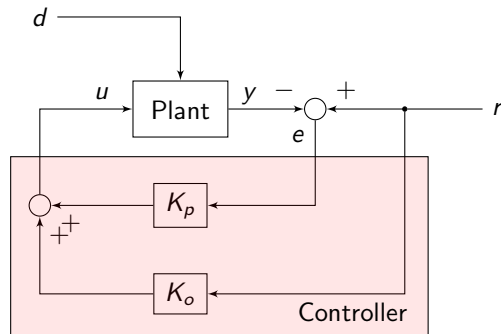
Proportional Controller: Simulation

- initial cond $y(0) = 35$, command $r = 65$ mph
- we try 4 different values $K_p = 0.1, 0.194, 0.25, q \dots$ can you guess the value of q ?
- notice as K_p increases step response gets more aggressive

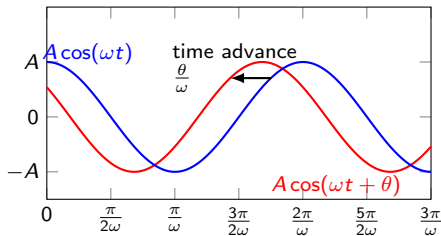
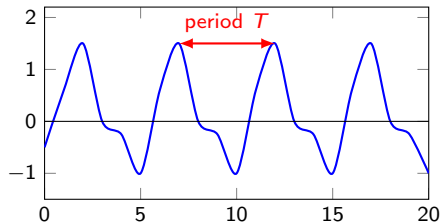


Proportional Control: Summary

- Pretty useful!
- Controller: $u = K_p(r - y) + K_o r$
- Feed-forward gain K_o chosen to get perfect tracking with no disturbances
i.e. DC gain from r to $y = 1$
- Feedback gain K_p chosen to set time constant i.e. how quickly $y(t)$ converges to r
- Aggressive controller: large K_p , quick response, small settling time
- If you get too aggressive, can get oscillations or instability on true plant model
we will see this later!



Periodic Signals and Sinusoids



- Period T , frequency $1/T$ Hertz, angular frequency $\omega = 2\pi/T$ rad/sec
- Standard form sinusoid $A \cos(\omega t + \theta)$
- Read off amplitude and frequency from plots
- Understand difference between time shift and phase shift
 - positive phase shift, advances the sinusoid (i.e.; it comes earlier)

The Sinusoidal Response

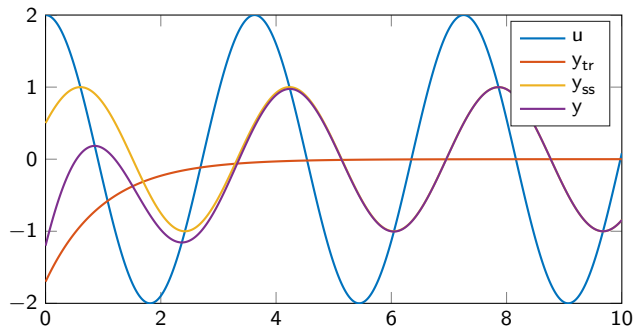
- Consider first-order model: $\dot{y} + ay = bu$
- Assume stable, i.e. free response decays to zero, or $a > 0$
- Apply a sinusoidal input

$$u(t) = \begin{cases} 0 & t < 0 \\ U \cos(\omega t) & t \geq 0 \end{cases}$$

- Then, the output is

$$y(t) = \underbrace{e^{-at}y_o - \frac{Uab}{a^2 + \omega^2}e^{-at}}_{y_{tr}} + \underbrace{\frac{Ub}{a^2 + \omega^2}(\omega \sin(\omega t) + a \cos(\omega t))}_{y_{ss}}$$

The Sinusoidal Response



The Frequency Response

- Consider stable first-order model: $\dot{y} + ay = bu$
- Sinusoidal input $u = U \cos(\omega t)$
- Steady-state response: $y_{ss} = \frac{Ub}{a^2 + \omega^2} (\omega \sin(\omega t) + a \cos(\omega t))$
- Can rewrite the steady-state term by trig identities:

$$y_{ss} = M \cdot U \cos(\omega t + \phi) \quad \text{sinusoid, same frequency as input!}$$

- amplification/attenuation $M(\omega)$, phase shift $\phi(\omega)$

	amplification	$M(\omega)$	$\frac{ b }{\sqrt{a^2 + \omega^2}}$	$\left \frac{b}{j\omega + a} \right $
– depend on input frequency ω	phase shift	$\phi(\omega)$	$-\tan^{-1}\left(\frac{\omega}{a}\right) + \angle(b)$	$\angle\left(\frac{b}{j\omega + a}\right)$

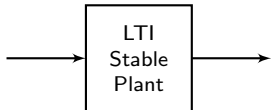
The Idea of Frequency Response

- Extremely important across all engineering:

Frequency Response

... if we apply a sinusoidal input of frequency ω to a *stable* first-order system, in steady-state the output will also be a sinusoid. It has the same frequency ω . However, it will be amplified or attenuated with a gain $M(\omega)$ and phase-shifted by $\phi(\omega)$.

input u
 $U \cos(\omega t)$
 sinusoid of
 frequency ω



output y
 $M \cdot U \cos(\omega t + \phi)$ + transient
 sinusoid of same frequency ω
 amplified/attenuated by $M(\omega)$
 phase shifted by $\phi(\omega)$

- Why is this important?

Bode Plots

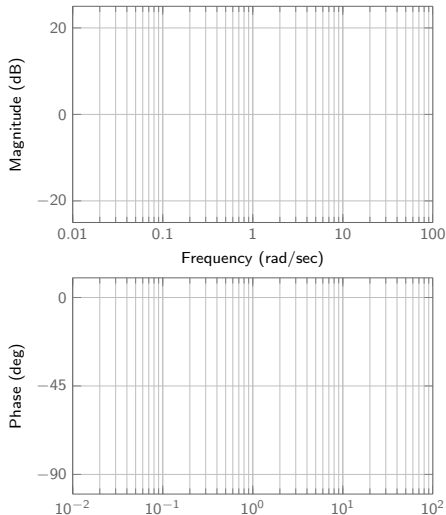
- Range of frequencies of interest very large
so we plot frequency on a log-scale
- Bode magnitude plot:
 - range of magnitudes is very large, so we use a log-scale
 - plot $20 \log_{10} M(\omega)$ vs. $\log_{10} \omega$
 - $M \leq 1 \iff M \leq 0\text{dB}$ $M \geq 1 \iff M \geq 0\text{dB}$
- Bode phase plot:
plot $\phi(\omega)$ vs. $\log_{10} \omega$

Decibels

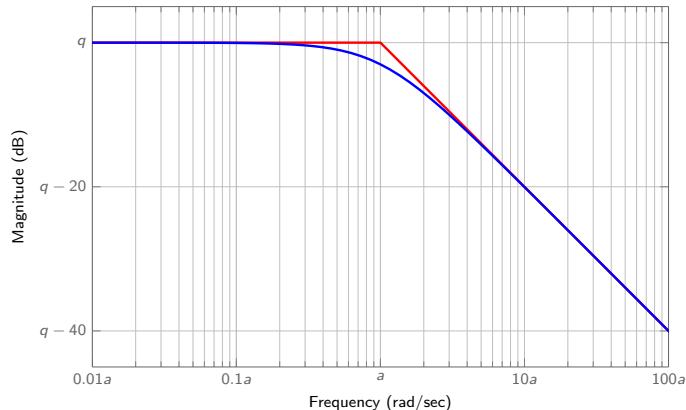
$M(\omega)$	$20 \log_{10} M(\omega)$
.01	−40 db
.1	−20 db
1	0 db
2	≈ 6 db
5	≈ 14 db
10	20 db
20	≈ 26 db
50	≈ 34 db
100	40 db

Ex 4: Bode Plots

- Consider stable first-order system $\dot{y} + 5y = 10u$
- Magnitude $M(\omega) = \left| \frac{10}{j\omega + 5} \right|$
 - large $\omega \gg 5$:
 $M(\omega) \approx 10/\omega = 20 \log_{10} 10 - 20 \log_{10} \omega$ dB
 straight line with slope -20 dB/decade
 - small $\omega \ll 5$:
 $M(\omega) \approx 10/5 = \text{DC gain} = 20 \log_{10} 2 = 6$ dB
 straight line with slope 0
- Phase $\phi(\omega) = \angle \left(\frac{10}{j\omega + 5} \right)$
 - large $\omega \gg 5$:
 $\phi(\omega) \approx \angle \left(\frac{10}{j\omega} \right) = -90^\circ$
 - small $\omega \ll 5$:
 $\phi(\omega) \approx \angle \left(\frac{10}{5} \right) = \angle \text{DC gain} = 0^\circ$

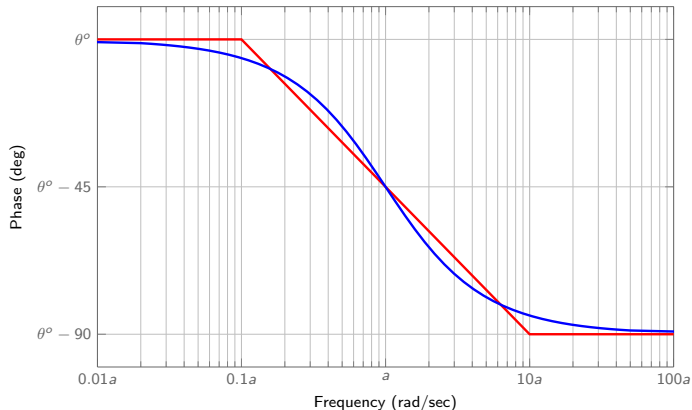


Bode Magnitude Plots



- exact plot in blue
- straight line approx in red
- start at $q = |\text{DC gain}| \text{ dB}$
 $q = 20 \log_{10} |b/a|$
- corner frequency = $|a| \text{ rad/sec}$
- slope after $a = -20 \text{ dB/decade}$
- decade = 10X increase in freq

Bode Phase Plots



- exact plot in blue
- straight line approx in red
- start at $\theta^o = \angle \text{DC gain}$
end at $\theta^o - 90^\circ$
- corner frequency = $|a|$ rad/sec
- slope from $0.1a$ to $10a = \pi/4$ rad/decade
decade = 10X increase in freq

Ex 5

Consider the first order LTI system $\dot{y} + ay = bu$. Suppose $a > 0$ and $b > 0$.

We observe two facts:

- in steady-state, the input $u(t) = \sin(t)$ is amplified by a factor of $\sqrt{2}$
- in steady-state, the input $u(t) = \sin(2t)$ is amplified by a factor of 1

Find a and b .

Ex 6

Sketch the frequency response of the first-order system $\dot{y} + 10y = -2u$.

